

4-1 · Understand Percent

Connect & apply · 8 tasks · Prepares for 6.RP.A.3.c

Use the explanation to discuss strategies. Accept mathematically equivalent answers.

1. FOUNDATION 1

Write thirty-seven hundredths as a fraction.

37/100

Hundredths means 100 equal parts; 37 are selected.

Skill: Read and write a fraction with denominator 100

2. FOUNDATION 2

A 10-by-10 grid has 4 complete rows shaded. How many squares are shaded?

40

Each row contains 10 squares; $4 \times 10 = 40$.

Skill: Count shaded squares on a 100-square grid

3. FOUNDATION 3

Write 68/100 as a decimal.

0.68

Put 68 in the hundredths places.

Skill: Write hundredths as a decimal

4. FOUNDATION 4

Simplify 45/100.

9/20

Divide the numerator and denominator by their common factor 5.

Skill: Simplify a fraction over 100

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5. CHECK A

37 of 100 squares are shaded — write it as a fraction, a decimal, and a percent.

Fraction: $37/100$. Decimal: 0.37. Percent: 37%.

Compare each part of your work with the worked solution.

6. CHECK B

62 of 100 squares are shaded — write it as a fraction, a decimal, and a percent.

Fraction $62/100$ (or $31/50$). Decimal 0.62. Percent 62%.

Compare each part of your work with the worked solution.

7. CHECK C

8 of 100 squares are shaded — write it as a fraction, a decimal, and a percent.

Fraction $8/100$ (or $2/25$). Decimal 0.08. Percent 8%.

Compare each part of your work with the worked solution.

8. TRANSFER

What does 150% look like on hundredths grids, and what decimal is it?

One fully shaded grid plus 50 squares of a second grid — $150/100 = 1.5$.

Explain why your method works and include units when needed.

4-2 · Relate Fractions, Decimals, and Percentages

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1. FOUNDATION 1

Compute 0.36×100 .

36

Multiplying by 100 makes each digit worth 100 times as much.

Skill: Multiply and divide by 100 by shifting the decimal point

2. FOUNDATION 2

Write $\frac{3}{8}$ as a decimal.

0.375

Divide 3 by 8; check $0.375 \times 8 = 3$.

Skill: Write a fraction as a decimal by dividing

3. FOUNDATION 3

Write $\frac{3}{4}$ as a percent.

75%

Multiply numerator and denominator by 25 to get $\frac{75}{100}$.

Skill: Recall benchmark equivalents for $\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{5}$, $\frac{3}{4}$, $\frac{1}{10}$

4. FOUNDATION 4

Which is greater: 0.6 or 0.56?

0.6

Write 0.6 as 0.60; 60 hundredths is greater than 56 hundredths.

Skill: Compare decimals to the hundredths place

4-2 · Relate Fractions, Decimals, and Percentages

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5. CHECK A

0.45 as a percent? 25% as a fraction in simplest form? Which is greater, 0.6 or 55%?

0.45 = 45%. 25% in simplest form: $25/100 = 1/4$. Comparison: $0.6 = 60\%$, so 0.6 is greater than 55%.

Compare each part of your work with the worked solution.

6. CHECK B

0.08 as a percent? 40% as a fraction in simplest form? Which is greater, 0.7 or 65%?

0.08 = 8%. 40% = $40/100 = 2/5$. 0.7 = 70%, so 0.7 is greater than 65%.

Compare each part of your work with the worked solution.

7. CHECK C

0.9 as a percent? 75% as a fraction in simplest form? Which is greater, 0.3 or 35%?

0.9 = 90%. 75% = $75/100 = 3/4$. 35% = 0.35, so 35% is greater than 0.3.

Compare each part of your work with the worked solution.

8. TRANSFER

Order least to greatest: 0.45, $2/5$, 48%, $1/2$.

As percents: 45%, 40%, 48%, 50%. Order: $2/5$, 0.45, 48%, $1/2$.

Explain why your method works and include units when needed.

4-3 · Estimate the Percent of a Number

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Use the explanation to discuss strategies. Accept mathematically equivalent answers.

1. FOUNDATION 1

Find 10% of 350.

35

Ten percent is one tenth; $350 \div 10 = 35$.

Skill: Find 10% of a number by shifting the decimal one place left

2. FOUNDATION 2

Find 1% of 800.

8

One percent is one hundredth; $800 \div 100 = 8$.

Skill: Find 1% of a number by shifting two places left

3. FOUNDATION 3

Find half of 86.

43

Halve 80 and 6 separately: $40 + 3 = 43$.

Skill: Halve and double a number mentally

4. FOUNDATION 4

Estimate 25% of 198 by using 200 as a compatible number.

50

25% is one fourth; $200 \div 4 = 50$.

Skill: Round to a compatible number that is easy to take a percent of

4-3 · Estimate the Percent of a Number

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5. CHECK A

10% of 320? 1% of 320? Half of 32?

**10% of 320 = 32 (shift decimal 1 place left).
1% of 320 = 3.2 (shift decimal 2 places left).
Half of 32 = 16 (which represents 5% of 320).**

Compare each part of your work with the worked solution.

6. CHECK B

10% of 480? 1% of 480? Half of 48?

10% of 480 = 48. 1% of 480 = 4.8. Half of 48 = 24, which is 5% of 480.

Compare each part of your work with the worked solution.

7. CHECK C

10% of 250? 1% of 250? Half of 25?

10% of 250 = 25. 1% of 250 = 2.5. Half of 25 = 12.5, which is 5% of 250.

Compare each part of your work with the worked solution.

8. TRANSFER

Estimate a 15% tip on a \$42.80 bill using benchmarks.

Round to \$42. 10% $\approx$ \$4.20, 5% $\approx$ \$2.10, so 15% $\approx$ \$6.30.

Explain why your method works and include units when needed.

4-4 · Find and Compare with Percentages

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1. FOUNDATION 1

Write 18% as a decimal.

0.18

Percent means per hundred; $18 \div 100 = 0.18$.

Skill: Write a percent as a decimal

2. FOUNDATION 2

Compute 60×0.35 .

21

$60 \times 35 \div 100 = 2100 \div 100 = 21$.

Skill: Multiply a whole number by a decimal

3. FOUNDATION 3

Find $\frac{3}{4}$ of 28.

21

Divide 28 by 4, then multiply by 3: $7 \times 3 = 21$.

Skill: Multiply a whole number by a fraction

4. FOUNDATION 4

Which is greater: 20% of 90 or 30% of 70?

30% of 70

The products are 18 and 21; 21 is greater.

Skill: Compare two products to decide which is larger

4-4 · Find and Compare with Percentages

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Use the explanation to discuss strategies. Accept mathematically equivalent answers.

5. CHECK A

0.25×80 ? $\frac{3}{4}$ of 80? Which is more: 20% of 50 or 30% of 40?

$0.25 \times 80 = 20$. $\frac{3}{4}$ of 80 = 60. Comparison: 20% of 50 = 10; 30% of 40 = 12. 30% of 40 is more ($12 > 10$).

Compare each part of your work with the worked solution.

6. CHECK B

0.4×90 ? $\frac{2}{5}$ of 90? Which is more: 30% of 60 or 25% of 80?

$0.4 \times 90 = 36$. $\frac{2}{5}$ of 90 = 36. 30% of 60 = 18; 25% of 80 = 20. So 25% of 80 is more.

Compare each part of your work with the worked solution.

7. CHECK C

0.6×50 ? $\frac{3}{5}$ of 50? Which is more: 40% of 70 or 50% of 60?

$0.6 \times 50 = 30$. $\frac{3}{5}$ of 50 = 30. 40% of 70 = 28; 50% of 60 = 30. So 50% of 60 is more.

Compare each part of your work with the worked solution.

8. TRANSFER

A \$60 jacket is 30% off. What is the sale price?

Discount: $0.30 \times 60 = \$18$. Sale price: $60 - 18 = \$42$ (or 70% of 60 = \$42).

Explain why your method works and include units when needed.

4-5 · Determine the Whole Given the Part and Percent

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Use the explanation to discuss strategies. Accept mathematically equivalent answers.

1. FOUNDATION 1

Compute $18 \div 0.3$.

60

Scale dividend and divisor by 10: $180 \div 3 = 60$.

Skill: Divide a whole number by a decimal

2. FOUNDATION 2

Twelve students out of a class of 30 walk to school. Identify the part and the whole.

Part 12; whole 30

The part counts walkers; the whole counts everyone in the class.

Skill: Tell which number is the part and which is the whole

3. FOUNDATION 3

Compute $9 \div \frac{1}{4}$.

36

There are 4 fourths in each whole, so $9 \times 4 = 36$ fourths.

Skill: Divide by a benchmark fraction such as $\frac{1}{4}$ or $\frac{1}{2}$

4. FOUNDATION 4

A student says the whole is 80 when 25% is 20. Check by finding 25% of 80.

20

$0.25 \times 80 = 20$, so the proposed whole checks.

Skill: Check an answer by multiplying back

4-5 · Determine the Whole Given the Part and Percent

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5. CHECK A

$12 \div 0.25$? 12 is 25% of what number? Now check: 25% of your answer.

$12 \div 0.25 = 48$. The whole is 48 (12 is 25% of 48). Check: 25% of 48 = $0.25 \times 48 = 12$ (balances!).

Compare each part of your work with the worked solution.

6. CHECK B

$15 \div 0.5$? 15 is 50% of what number? Now check: 50% of your answer.

$15 \div 0.5 = 30$. The whole is 30. Check: $0.5 \times 30 = 15$. ✓

Compare each part of your work with the worked solution.

7. CHECK C

$9 \div 0.2$? 9 is 20% of what number? Now check: 20% of your answer.

$9 \div 0.2 = 45$. The whole is 45. Check: $0.2 \times 45 = 9$. ✓

Compare each part of your work with the worked solution.

8. TRANSFER

A team won 18 games, which was 75% of the games played. How many games were played?

$18 \div 0.75 = 24$ games. Check: $0.75 \times 24 = 18$. ✓

Explain why your method works and include units when needed.